

Bulk-color mechanism v0.2 — Direction (A) sharpened: COLOR = $SO(3) \subset SO(5)$ sub-vector 3-fold in the BULK holomorphic tangent. The 5-vector of $SO(5)$ decomposes under $SO(3) \times SO(2)$ as $3 + 2$ — the 3 is a canonical bulk-resident 3-fold structure that $K=SO(5) \times SO(2)$ carries but doesn't expose. Closes Keeper's two-structures burden: color from $SO(3)$ sub-vector (geometric); generations from spinor/ $h^\vee$ tower (algebraic). Two STRUCTURALLY DISTINCT 3-folds, both = 3 from $B_2/SO(5)$ data.

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0. Where this sits

The v0.1 mapping laid out three candidate directions for the bulk-color mechanism. This v0.2 PUSHES Direction (A) — bulk K-type structure — concretely, and finds a specific candidate: the $SO(3) \subset SO(5)$ sub-vector 3-fold in the bulk holomorphic tangent.

1. The candidate (concrete structural identification)

$SO(5) \supset SO(3) \times SO(2)$ branching of the 5-vector (rigorous, standard):

$$5 \text{ (vector of } SO(5)) \downarrow SO(3) \times SO(2) = (3, 0) + (1, \pm 1)$$

The 5-dim $SO(5)$ vector rep decomposes under $SO(3) \times SO(2)$ as a 3-dim $SO(3)$ vector (spatial 3-vector) plus a 2-dim $SO(2)$ charge ± 1 doublet. The 3-dim piece is the rotational/spatial vector inside the $SO(5)$ tangent; the 2-dim piece carries the $SO(2)/U(1)$ charge structure.

The candidate:

COLOR (3-fold) = the $SO(3)$ sub-vector inside the $SO(5)$ bulk holomorphic tangent $T^\wedge(1,0) = 5_+(+1)$.

Each “color” of a quark = one of the 3 directions in the $SO(3)$ -vector subspace of the 5-dim bulk tangent. The 3-fold is geometric ($SO(3)$ rotation of the 5-dim bulk), residing in the BULK (not in K 's standard branching of the 5-vector under $SO(5) \times SO(2)$).

2. Why this is in the bulk, not in K

$K = SO(5) \times SO(2)$ acts on the 5-vector as the full $SO(5)$ rotation. The $SO(3) \times SO(2)$ decomposition is a FURTHER refinement that K 's standard K-type branching doesn't surface — it's a sub-structural decomposition of the $SO(5)$

representation. The $SO(3)$ sub-vector is “visible” only when you branch further inside $SO(5)$.

The bulk of D_{IV}^5 has access to this sub-structure because the bulk’s geometric content includes the full $SO(5)$ action on the 5-vector tangent, which can be refined to $SO(3) \times SO(2)$. The Shilov boundary, by contrast, has stabilizer $SO(4) = SU(2) \times SU(2)$, which doesn’t naturally contain $SO(3) \times SO(2)$ ($SO(4)$ and $SO(3) \times SO(2)$ are different rank-2 subgroups of $SO(5)$).

So: - Shilov boundary: stabilizer $SO(4) = SU(2)_L \times SU(2)_R \rightarrow$ carries the electroweak $SU(2)_L$. NO 3-fold color structure ($SO(4)$ doesn’t contain $SO(3) \times SO(2)$). - Bulk: full $SO(5)$ available, refines to $SO(3) \times SO(2) \rightarrow$ carries the 3-fold $SO(3)$ sub-vector = color.

This is consistent with: leptons (Shilov) are colorless (no $SO(3)$ sub-vector in the Shilov stabilizer), quarks (bulk) are colored (3-fold $SO(3)$ sub-vector accessible in the bulk tangent).

3. Closing Keeper’s two-structures burden

Keeper’s burden on route (II): “one $h^{\vee}=3$ doing double duty” needs to derive TWO INDEPENDENT 3-fold structures (3 colors $\times$ 3 generations = 9 quark combinations). The v0.2 candidate addresses this:

structure	source	character
COLOR (3)	$SO(3) \subset SO(5)$ sub-vector in bulk tangent	GEOMETRIC — branching of the 5-vector under $SO(3) \times SO(2)$
GENERATIONS (3)	$h^{\vee} = N_c$ spinor-tower (route II, with #414 candidate mechanism)	ALGEBRAIC — Coxeter/Hall-algebra structure

These are STRUCTURALLY DISTINCT 3-folds: color is geometric (rotational sub-vector), generations are algebraic (spinor/Coxeter tower). They are INDEPENDENT (different mathematical objects). Both equal 3 because both follow from $B_2/SO(5)$ structure, but the routes are different.

Keeper calibration HELD: TWO B_2 -specific 3-folds coinciding numerically — NOT “one $h^{\vee}$ doing double duty.” The two are independent in mechanism, share only the B_2 structure they’re both rooted in.

The cross-cutting nature falls out cleanly: 3 colors (geometric, in bulk tangent) $\times$ 3 generations (algebraic, in spinor tower) = 9 quark combinations naturally, because the two structures act on different aspects of the quark state (its bulk tangent direction $\times$ its spinor-tower index).

4. The CON (honest)

$SO(3) \neq SU(3)$: standard QCD color is $SU(3)$ gauge symmetry (dim 8), not $SO(3)$ rotation (dim 3). If color = $SO(3)$ sub-vector, the substrate’s color structure is NOT a gauge group in the strict QCD sense — it’s a COUNTING structure (3 directions in a sub-vector) that must reproduce $SU(3)$ gauge phenomenology effectively at low energy.

This is consistent with the counting-not-symmetry hypothesis (Casey’s Track P / route II direction): the substrate’s “3” comes from a structural counting (3 $SO(3)$ directions or 3 $h^{\vee}$ levels), and the $SU(3)$ gauge phenomenology of low-energy QCD is EMERGENT, not fundamental.

Burden remaining: demonstrate that 3 substrate-resident color directions reproduce $SU(3)$ low-energy QCD phenomenology — confinement, asymptotic freedom, the gluon octet (8 gluons of $SU(3)$). The mechanism for 8 gluons from a 3-vector mechanism is not yet clear; this is the open work.

Possible direction: the gluon octet could be the $8 = \dim(SU(3))$ emerging from the substrate’s adjoint structure on the bulk — and the substrate’s “adjoint of color” might be the symmetric-traceless part of the $SO(3) \times SO(3)$ tensor product ($= 5 + 3 = 8$). I’ll flag this as a candidate-of-a-candidate, not a derivation.

5. Direction (A) tested cleanly; Directions (B) + (C) status

- Direction (A) — bulk K-type structure: SHARPENED to $SO(3) \subset SO(5)$ sub-vector. Candidate v0.2 (this doc). Clean structural identification; open burden on $SU(3)$ emergence.
- Direction (B) — spinor³ as bulk anchor: Elie's E7 has spinor³ multiplicity 3 (B_2 -specific). Per Keeper's calibration, this 3 = count of bosonic fundamentals of B_2 (trivial+vector+adjoint), NOT manifestly $h^\vee$ in origin — a numerical coincidence at B_2 . So Direction (B) doesn't cleanly tie to color via $h^\vee$. Possibly the 3 spinor³ channels label something else (e.g., the 3 “matter types” Elie suggested for generations). Leave Direction (B) for the generation mechanism (#414 v0.2 candidate), NOT for color.
- Direction (C) — $Z/3$ quotient probe: no clean $Z/3$ found in D_{IV}^5 's structure (rank=2, no $Z/3$ in the Weyl group D_4 of B_2 , Bergman exponent 5/2 doesn't yield modular $Z/3$ residues). Direction (C) was likely a dead end; Direction (A) is the better candidate.

6. Honest scope + tier

RIGOROUS (rep theory): $SO(5) \supset SO(3) \times SO(2)$; branching $5 = (3,0) + (1,\pm 1)$; $SO(3)$ is a 3-dim subspace of the 5-dim $SO(5)$ -vector; $SO(3)$ is a subgroup of $SO(5)$ (not of $K = SO(5) \times SO(2)$ in a way that surfaces in standard K-type branching).

CANDIDATE (this doc): COLOR = $SO(3)$ sub-vector of the bulk holomorphic tangent. Structurally clean, geometrically motivated, addresses two-structures burden honestly (color geometric, generations algebraic — independent mechanisms).

OPEN burden: demonstrate this 3-fold structure reproduces $SU(3)$ low-energy QCD gauge phenomenology (confinement, 8 gluons, asymptotic freedom). The counting-not-symmetry hypothesis says it should as an emergent effective theory; the explicit mechanism is the multi-week work.

Cal #27 / honesty: I am NOT claiming color $SU(3)$ is derived. I am proposing a CANDIDATE mechanism — color as the $SO(3) \subset SO(5)$ sub-vector 3-fold in the bulk tangent — that is structurally clean, in the right place (bulk, not K), and pairs with an algebraically-distinct generation mechanism ($h^\vee$ spinor-tower) to give two independent 3-folds (Keeper's burden addressed). The $SU(3) \neq SO(3)$ gap is the honest open burden on the candidate. Direction (A) sharpened from “research-staged” to “candidate mechanism.”

Routed: → Keeper: tier-gate this candidate; the two-structures burden is addressed STRUCTURALLY (color geometric vs generations algebraic), but the $SU(3) \neq SO(3)$ gap is the new burden. → Elie: $SO(5) \supset SO(3) \times SO(2)$ branching of higher K-types (Sym^k of 5-vector) — does the $SO(3)$ substructure persist and grow in the bulk K-type tower? Would test whether the 3-fold extends naturally to the full bulk. → Grace: catalog scan for “ $SO(3) \subset SO(5)$ structure” — does any known SM observable cleanly identify with the 3-direction sub-vector? → me: next push is on (a) the $SU(3)$ emergence from $SO(3)$ (multi-week, may need help), and (b) verifying the 3 colors $\times$ 3 generations independence concretely on a quark per-particle entry.

— Lyra, bulk-color v0.2 Direction (A) sharpened. CANDIDATE: COLOR = $SO(3) \subset SO(5)$ sub-vector 3-fold in the bulk holomorphic tangent $T^{(1,0)} = 5_{-} (+1)$, from the standard branching 5 of $SO(5) = (3, 0) + (1, \pm 1)$ under $SO(3) \times SO(2)$. 3-fold is GEOMETRIC, in the BULK (not K's standard branching). Closes Keeper's two-structures burden HONESTLY: color geometric ($SO(3)$ sub-vector) vs generations algebraic ($h^\vee$ spinor-tower) — STRUCTURALLY DISTINCT 3-folds, both = 3 from $B_2/SO(5)$ data. CON: $SO(3) \neq SU(3)$; QCD- $SU(3)$ gauge phenomenology must EMERGE from the 3-direction counting (Track P / counting-not-symmetry hypothesis). Open burden: derive $SU(3)$ emergent QCD from substrate $SO(3)$ counting. Multi-week, but the structural candidate is now in hand.